

BIF401 FINAL TERM PAST PAPER

BIF 401 (5PM)

Homology modeling

Loops characteristics

Chu fasman alogarithm

Ab initio model and limitations

How sequence of protein identified

What do u know about Ramachandran plot

Bioinformatics paper

Total qstn 47, mcqs are 32

Paper was easy

Q:need of chous fasmen method

Q: alphex helix regions in chou fasmen meethod

Q:formula to calculate tetra peptide

Q: why less energy used in folded rna as compare to unfloded

Q: role of mass spectrometry in protein

Q: domain in protein

Bif401 today paper

*what are conserved sequences 2

*Name three fragmentation techniques 2

*Deferentiate b/w Bottom up and top down proteomics 3

*What happen in Jackson traceback algorithm if two or more scoring matrices scores higher
construction ya range tha i think 3

*hydrophobicity of amino acid 3

*Enlist five Institutes for bioinformaticians all over the world 5

BIF 401 Paper Timing 8; 00 am

1. Hydrophobic amino acids?

2. optimal energy function in protein?

3. PH<PK?

4. Need for chou fasman algorithm?

5. Blast and Fasta tools in sequence alignmnedt and its differences?

6. Protein scoring?

7. Why is it difficult to predict the structure of protein?

8. Describe the following?

- a. role of amino acids.
- b. folded protein.
- c. denature protein
- 9. experimental techniques in bioinformatics?
- 10. Silico fragments comparison?
- 11. Describe PDB in the following.
 - a. TITLE;
 - b. SOURCE;

bif401 final term (saturday)
 choufasman algorithm??
 improvements of choufasman algorithm?
 limitations of choufasman?
 initio modeling??
 scoring formulas??
 ph>pk??
 beta sheet by choufasman??
 alpha helix by choufasman??

BIF 401 (5PM)
 Homology modeling
 Loops characteristics
 Chu fasman algorithm
 Ab initio model and limitations
 How sequence of protein identified
 What do u know about Ramachandran plot

.yesterday BIF 401 on 2;30 pm
 PAPER WAS ...
 Role of amino acids .2
 pH value less than pk .2
 Chao fasman ..5
 Ab anitio devileers ..2.
 Sillico alogrithm ..5
 Two groups of amino acids .2

Qno.1 what is principle of dynamic programming. 2
 Qno.2 what is role of scoring scheme in top down proteomics.2
 Qno.3 what happend when pH is less then pk value.2
 Qno.4 what are the basic need in Chou Fasman Algorithm.2

Qno.5 what is ab initio modelling?describe its limitations.2
 Qno.6 describe following elements ,title,source,RAVDAT,COMPND.2
 Qno.7 enlist famous protein sequence database.2
 Qno.8 write down characteristics of amino acids.3
 Qno.9 why is structure prediction of protein is carried out.3
 Qno.10 how many types of force involved in folding.3
 Qno.11 why may we need to model protein structure when we have X rays and NMR techniques.3
 Qno.12 why do hairpin loops reverse the chemical direction of molecules.3
 Qno.13 how protein sequence are identified.5
 Qno.14 what are the optimal energy function in protein structure prediction.5
 Qno.15 describe the following.5
 a.advantage of folded protein over denature protein
 b.factors affect in folding protein
 c.role of amino acid
 d.goal of folding protein

Role of amino acid

Folded of protein goal

Chou fasman

Why difficult to determine structure of protein

What is hydrophobicity of amino acids

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M.C.Q.S are so difficult related to the topics of chou fasman algorithm, fasta, protein amino acids, most were from final term syllabus

BIF 401 final paper today at 11:00 am

1.the term ExPANSy stands for?(2)

2. differentiate between bottom up proteomics and bottom down proteomics (3)

3. how loops are found after the finding of alpha and beta sheets (5)
4. how proteins sequences identified?(5)
5. explain all (5)
 - * role of amino acids.
 - * how folding of proteins are useful over the denature proteins
 - * why folding is important
 - * what factors are used for folding procees
6. why the tertiary structure of RNA is less stable over the secondary structure of RNA (5)
7. what is ab initio modelling describe the limitations of this model (3)
8. write down the following terms of PDB (3)
 - * title
 - *source
 - * COMPND
 - * REVEDAT
9. how we find the secondary structures of proteins by using a model?(3)
10. what happend when pH is less than the pK value (2)
11. define ionization in terms of proteins (2)
12. what you have a home message from this course (3BIF 401 final paper today at 11:00 am)

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Bif 401 time 2 30

- 1 Choufosman alpha or beta structure of 2ndry structure
- 2 system biology and application

3 if pH is less than pK then what happens
4 define following
Goals of folding
Factors in folding
Character of amino acid
5 scores in silico fragmentation
6 what is hydrophobicity of amino acid
7 what are characteristic of amino acid
Think

few mcqs of BIF 401
1. Domains are----
2. RNA brick database-----
3. overcome homology modelling limitations by-----
4. homology predicts structure of protein-----
5. aromatic amino acids-----
6. DNA/RNA sequence----database
7. mRNA does not have-----
8. BLAST can find sequences-----
9. MS based proteomics begins measurement of -----
10. alpha carbon atom can be obtained by-----

Bif401 15-aug-2017
Repeated questions
Protein sequence
Protein folding
Chou-Fasman
Ab initio
Protein folding
Propensity
Protein n sequence database name
pH less than pK
Fragmentation of protein k 2 question type

Bif401

Repeated questions

Protein sequence

Protein folding

Chu fousman

Ab nitio

Protein folding

Propensity

Protein n sequence database name

PH less than pk

Fragmentation of protein k 2 question thy |

bif401 final term (saturday)
choufasman algorithm??
improvements of choufasman algorithm?
limitations of choufasman?
initio modeling??
scoring formulas??
ph>pk??
beta sheet by choufasman??
alpha helix by choufasman?

BIF 401 Paper Timing 8; 00 am

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BIF401

Short questions

- 1) Define FASTA?
- 2) Define Clustering and explain two types of phylogenetic tree?
- 3) What happen when pH is less than pK?
- 4) Compare tBLASTn and tBLASTx?
- 5) Explain ionization with reference to proten?
- 6) Charecteristics of amino acid?
- 7) Define Ab initio modeling and its limitations?
- 8) TITLE SOURCE REVDAT and COMPOUND
- 9) Why amino acid are hydrophobic in nature?
- 10) Enlist any 5 fields which employed the bioinformatics?

Long Questions

- 1) Explain Chau fasman for alpha helix nd beta turn?
- 2) Protein sequence identification?
- 3) Explain following Folding of protein / denatured proteins advantages

Role of amino acid

Tool used in protein folding Goal of protein folding

15aug2017

Bif401 today paper

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Bif 401 today 8 bje5 16aug2017

types of RNA and functions.

Goals of folding identification of protein 10.

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Q. how proteins sequences identified? (5)

5. explain all (5) *

role of amino acids. *

how folding of proteins are useful over the denature proteins

* why folding is important

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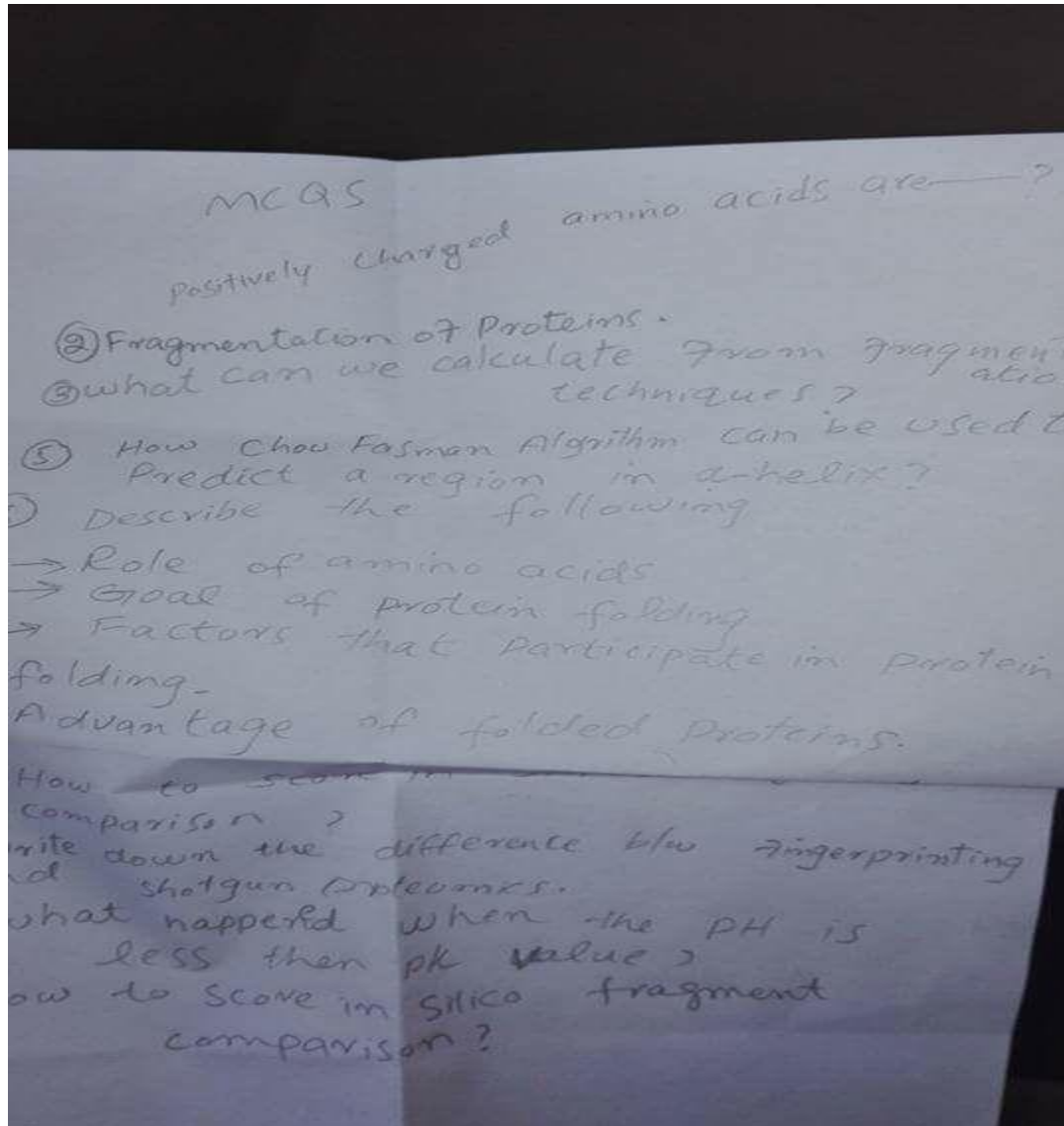
7 what are characteristic of amino acid

Orfs of DNA

Bonding between the secondary structures of protein

Bif 401 today 8 bje5 16aug2017

types of RNA and functions. Goals of folding identification of protein 10. what happens when pH is less than the pK value (2) . what is ab initio modelling describe the limitations of this model (3) 8. write down the following terms of PDB (3)* title*source * COMPND* REVEDAT Q.how loops are found after the finding of alpha and beta sheets (5) Q. how proteins sequences identified?(5) 5. explain all (5) * role of amino acids. * how folding of proteins are useful over the denature proteins * why folding is important Q: role of loops in reverse of chemical reaction



Role of amino acids .2
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- 1- $pH < pK$
2. Chou Fasman algorithm
3. Ab initio
4. In silico
5. hydrophobicity of amino acid.
6. Role of amino acid.
7. Wobble Test hypothesis.
8. Difference b/w FASTA & BLAST.
9. Protein folding.
10. Factors of folding (protein)
11. Goal of folding.
12. PDB file format.
 - * title
 - * source
 - * Compound
 - * Revedat.
13. Advantage of folded protein than denatured protein.
14. TDP.

MCQs mostly from final, few from Protein & rest of other from different algorithms. (Total - 32 MCQs).